Calendar
Combinatorics
Fall 2026

I will put all deadlines on the Blackboard Calendar. Please check it regularly.

0.1 Key university dates

For official University dates, see the Fall 2026 academic calendar and the Registrar's final-exam schedule.

Date	Event
Monday, August 17	Classes begin
Monday, September 7	Labor Day holiday
Tuesday, November 24–Friday, November 27	Thanksgiving holiday
Wednesday, December 9	Final exam: Wednesday, December 9, 2026, 8:00-10:00

0.2 Exams

Date	Assessment
Friday, September 18	Midterm 1
Friday, October 16 TENTATIVE	Midterm 2
Friday, November 13 TENTATIVE	Midterm 3
Wednesday, December 9, 2026, 8:00-10:00	Exam 4 and mastery reassessment

0.3 Daily lecture log

The public daily log below contains the lectures currently released. Later planning entries remain in the instructor's private lecture log and do not appear in this navigation.

Wk	Lec	
		L1: Set notation and basic counting
1	2026-08-17	Sets, cardinality, Cartesian products, and the sum rule.
		L2: Product rule and strings
	2026-08-19	Product rule; strings with and without repetition.
		L3: More counting techniques
	2026-08-21	Subtraction, division, permutations, combinations, and counting models.
		L4: Applications of combinations and combinations with repetition
2	2026-08-24	Lattice paths. Repeated-letter anagrams and multinomial coefficients. Combinations with r
		L5: Review of choice models, increasing sequences, and basic binomial expansio
	2026-08-26	Choosing the correct model: does order matter and are repetitions allowed. A second bijec
		L6: Polynomials, paths, Pascal recursion and restricted lattice paths, intro to in
	2026-08-28	Multinomial formula and polynomial expansions Coefficient extraction from powers of mult
		L7: Catalan paths and the inclusion–exclusion principle
3	2026-08-31	Reflection argument and Catalan numbers. Inclusion–exclusion for two, three, and finitely r
		L8: Applications of inclusion–exclusion
	2026-09-02	Characteristic functions. Conclusion of the proof of the inclusion–exclusion principle. Comp
		L9: Applications of inclusion–exclusion
	2026-09-04	Applications to bounded integer solutions and strings without repeated letters that avoid o
		L10: Applications of inclusion–exclusion
4	2026-09-09	Review of counting strings that avoid specified substrings. The inclusion–exclusion formula
		L11: Applications of inclusion–exclusion
	2026-09-11	Strings that use every symbol, surjective functions, and Stirling numbers. Graphs without i
		L12: Rook polynomials and forbidden positions
5	2026-09-14	Review of rook numbers and rook polynomials, including examples of their computation an
		L13: Midterm 1
	2026-09-18	Midterm 1.
		L14: Ordinary generating functions: definitions and series multiplication
6	2026-09-21	Ordinary generating functions; encoding finite and unlimited selections; substitution to trac
		L15: Ordinary generating functions: advanced techniques
	2026-09-23	Infinite formal power series and rational functions; constrained integer equations; combinat
		L16: TODO: Friday 9/25
	2026-09-25 TENTATIVE	
		L17: TODO: Monday 9/28
	2026-09-28 TENTATIVE	
		L18: TODO: Wednesday 9/30
	2026-09-30 TENTATIVE	
		L19: Solving homogeneous linear recurrences with constant coefficients
	2026-10-02	Introduction to solving homogeneous linear recurrences with constant coefficients: classifica
		No class
7	2026-10-05	No class.
		L20: Characteristic polynomials, repeated roots, and nonhomogeneous recurren
	2026-10-07	The characteristic polynomial of a linear recurrence with constant coefficients. Solutions of
		L21: TODO: Friday 10/9
	2026-10-09 TENTATIVE	
		L22: TODO: Monday 10/12
	2026-10-12 TENTATIVE	
		L23: Finite state machines 2: recurrence relations and fast computation
	2026-10-14	From a transition-matrix (vector) recurrence to a scalar recurrence relation. Computing ter
		L24: Midterm 2
	2026-10-16	Midterm 2.
		Fall break
8	2026-10-19	Fall break; no class.

TENTATIVE Lecture topics and assessment dates may change.

0.4 Week and unit overview

Weeks	Unit	Topic
1–4	Unit I: Enumeration	Counting models, bijections, double counting, and incl
4–5	Unit II: Generating-function foundations	Formal power series and ordinary generating-function
6–8 TENTATIVE	Unit III: Recurrences	State models, linear recurrences, recursive evaluation,
9–16 TENTATIVE	Further topics	To be announced